Predicting NHL Game Totals

NHL Betting Analytics Pipeline

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Synopsis

This project investigates whether pre-game team performance indicators can be used to forecast NHL game totals in a way that supports better over/under decision-making. Using a dataset of 22,118 games and 473 engineered numeric features derived exclusively from MoneyPuck statistics, the analysis finds that NHL scoring environments are neither random nor fully captured by simple averages. Total goals are strongly shaped by pace, shot volume, shot quality, and team style. A linear regression model generalizes well to unseen games (Test MAE = 0.606, Test R^2 = 0.892) and outperforms a more flexible XGBoost model that exhibits substantial overfitting. In addition, teams separate cleanly into high-event and low-event style archetypes, providing interpretable context for matchup-driven totals analysis. Together, these results suggest that disciplined, structure-aware modeling can form the foundation of a practical NHL totals forecasting framework, while also highlighting important limitations that must be addressed before real-world deployment.

1. Introduction

1.1 Overarching Question

The central question guiding this capstone is whether pre-game information can be used to identify NHL games that are structurally more likely to be high-scoring or low-scoring, and whether those insights can support improved decision-making in totals (over/under) betting markets. Rather than attempting to predict exact scores or game outcomes, this project focuses on forecasting the overall scoring environment of a game, which maps directly to totals wagering decisions.

1.2 Background and Motivation

Totals markets in professional sports are widely viewed as efficient, particularly in mature leagues such as the NHL. Goals are discrete events, the scoring range is bounded, and sportsbooks incorporate large volumes of historical data when setting lines. As a result, naive approaches based on season averages or recent box score outcomes tend to fail. However, market efficiency does not imply the absence of structure. Differences in team pace, shot quality, and defensive suppression create scoring environments that vary meaningfully from game to game.

Advanced hockey analytics aim to measure the process that generates scoring rather than outcomes alone. Expected goals (xG), for example, assigns probabilities to unblocked shot attempts based on shot location, type, and context, providing a principled measure of chance quality that is less noisy than raw goal counts (MoneyPuck, 2024). League-level scoring has also changed substantially over time, making temporal awareness essential when training and evaluating predictive models (Hockey-Reference, 2024). From a stakeholder perspective, even small, repeatable improvements in estimating expected totals can be valuable when applied consistently across a large number of games.

1.3 Hypothesis and Prediction

The working hypothesis is that recent team performance captured through rolling and exponentially weighted indicators of offense, defense, pace, and shot quality contains predictive signal about game scoring environments that is not fully reflected in simple historical averages. The associated prediction is that models trained on these features will generalize to future games and that teams will separate into distinct style archetypes associated with systematically higher or lower total goals.

2. Data

2.1 Data Acquisition

All data used in this project are derived from MoneyPuck, a public hockey analytics resource that provides game-level and team-level metrics based on shot quality and expected goals. No sportsbook pricing or betting-line data are included in the final version of this analysis. Restricting the data source to MoneyPuck ensures consistency in feature definitions and avoids complications related to market behavior or line movement.

2.2 Dataset Summary

The finalized dataset contains 22,118 NHL games spanning multiple seasons. In total, the dataset includes 487 columns, of which 473 numeric features were retained for modeling after exclusions and cleaning. The response variable, total_goals, has the following distribution:

- Count: 22,118
- Mean: 5.72 goals
- Standard deviation: 2.31 goals
- Minimum: 0 goals
- Maximum: 17 goals

Approximately 62.1% of games fall between 4 and 7 total goals, while 20.0% of games reach 8 or more goals, highlighting both the bounded nature of scoring and the presence of a meaningful right tail.

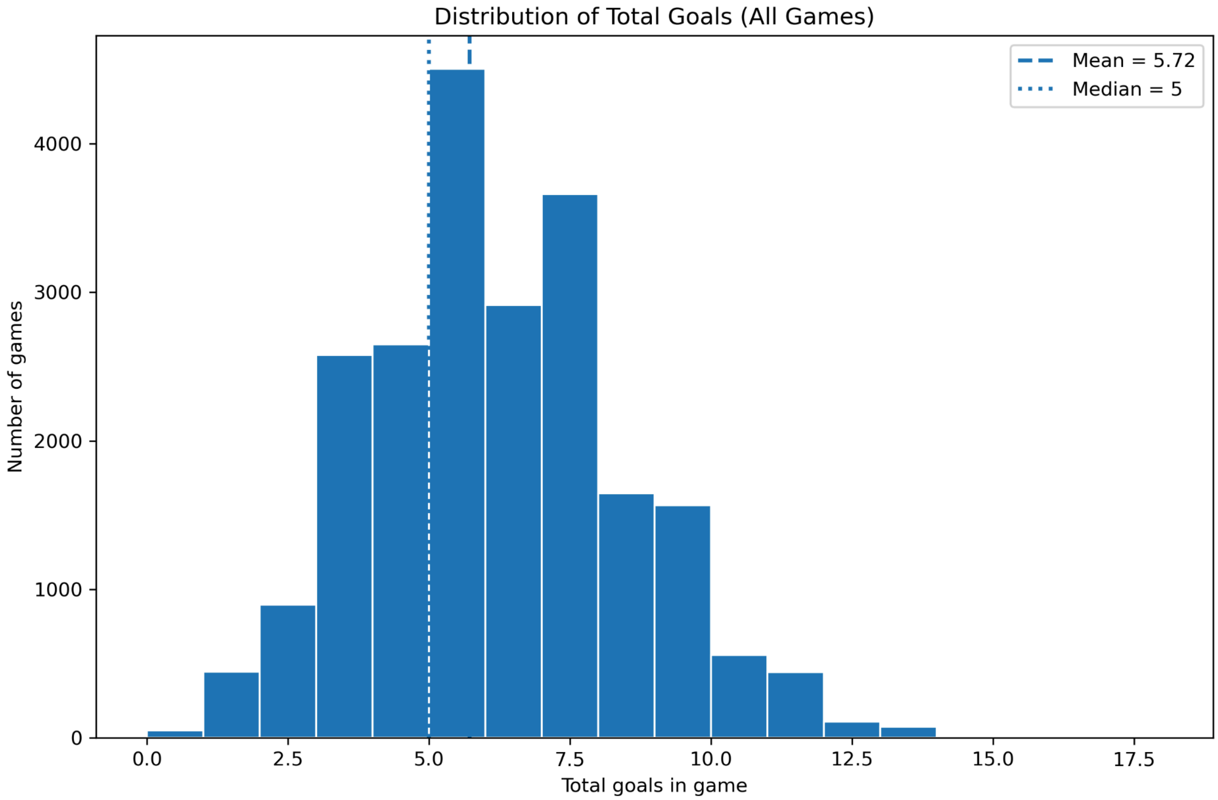

Figure 1. Distribution of total goals scored per game, with mean and median markers.

2.3 Response Variable

The response variable is total_goals, defined as the combined number of goals scored by both teams in a game. This target was chosen deliberately because it maps directly to over/under betting markets and avoids confounding team strength with scoring environment. Errors in predicting this variable correspond directly to errors a bettor would experience when wagering on totals.

2.4 Data Cleaning and Preparation

Data cleaning prioritized reproducibility and temporal integrity. Team identifiers were standardized, games were ordered chronologically by date, and rolling and exponentially weighted features were computed using only past information to avoid look-ahead bias. Columns that were entirely missing or constant were removed. Remaining missing values introduced primarily through rolling-window construction were handled using median imputation, a conservative choice that preserves central tendency while limiting sensitivity to outliers.

3. Exploratory Data Analysis

Exploratory data analysis was conducted to understand the distribution of scoring, assess temporal trends, and examine how process-based metrics relate to realized totals. Five figures summarize the most relevant characteristics of the data.

3.1 Distribution of Total Goals

Figure 1 shows that the distribution of total goals is centered at a median of 5 goals, with the bin corresponding to five goals clearly the tallest. Counts decline immediately above this value before increasing again, indicating mild multimodality rather than a smooth bell-shaped distribution. The right tail is present but lighter than the left tail, suggesting that extremely high-scoring games are less common than very low-scoring games. A noticeable spike appears at 7.5 goals, consistent with discrete scoring outcomes clustering around common totals.

3.2 Temporal Trends in Scoring

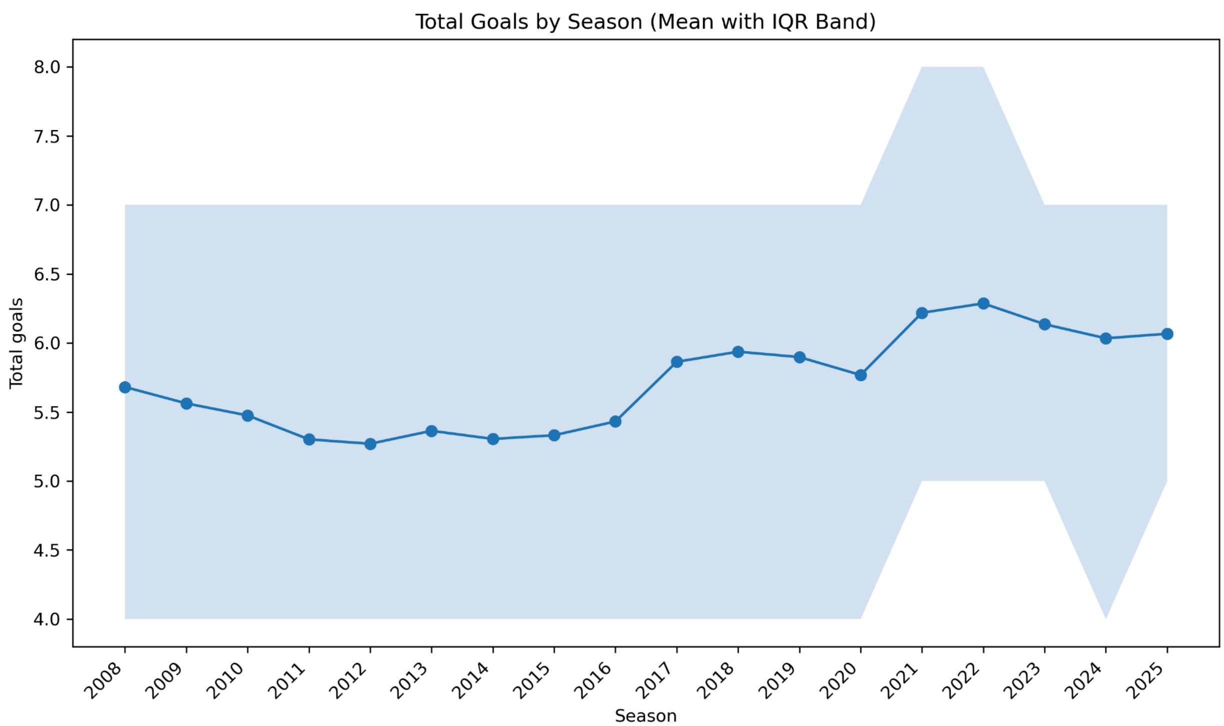

Figure 2. Mean total goals per game by season with interquartile range bands.

Mean scoring levels vary by season. The lowest-scoring season in the dataset is 2012, with an average of 5.27 goals per game, while the highest-scoring season is 2022, with an average of 6.29 goals per game which is a difference of just over 1.0 goal per game. Scoring trends upward from the early 2010s and appears to flatten after approximately 2021. The interquartile range changes noticeably during the 2021–2023 period, indicating an increase in game-level scoring. These patterns motivate the use of time-aware evaluation rather than random train–test splits.

3.3 Total Goals and Expected Scoring Environment

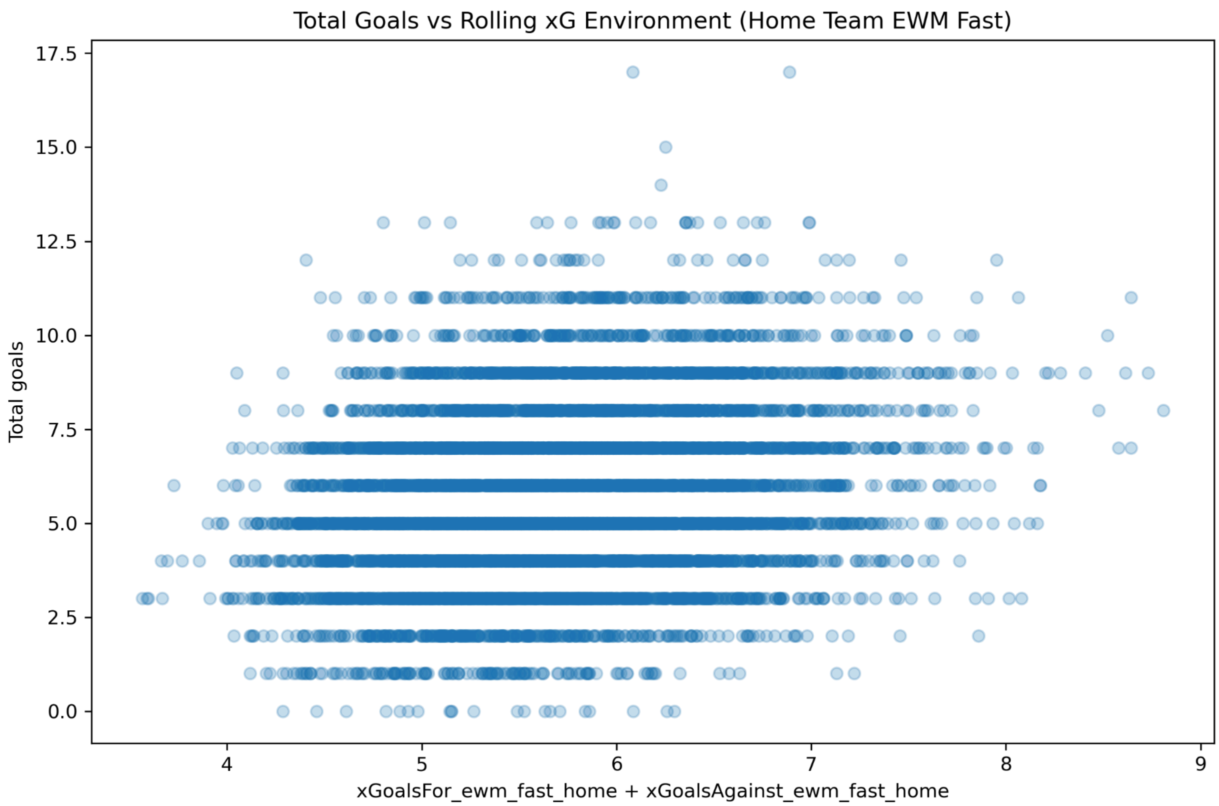

Figure 3. Total goals versus rolling expected-goals environment (home team xGoalsFor + xGoalsAgainst, fast EWM).

Figure 3 compares realized total goals to a rolling expected-goals environment. The relationship is positive but weak, with a Pearson correlation of 0.27, and substantial dispersion around the trend. Most observations cluster between 4 and 7 goals, while outliers appear on all sides of the distribution. This behavior reinforces that expected goals describe the underlying scoring process rather than guaranteeing outcomes at the individual game level.

3.4 Total Goals and Rolling Goal Differential

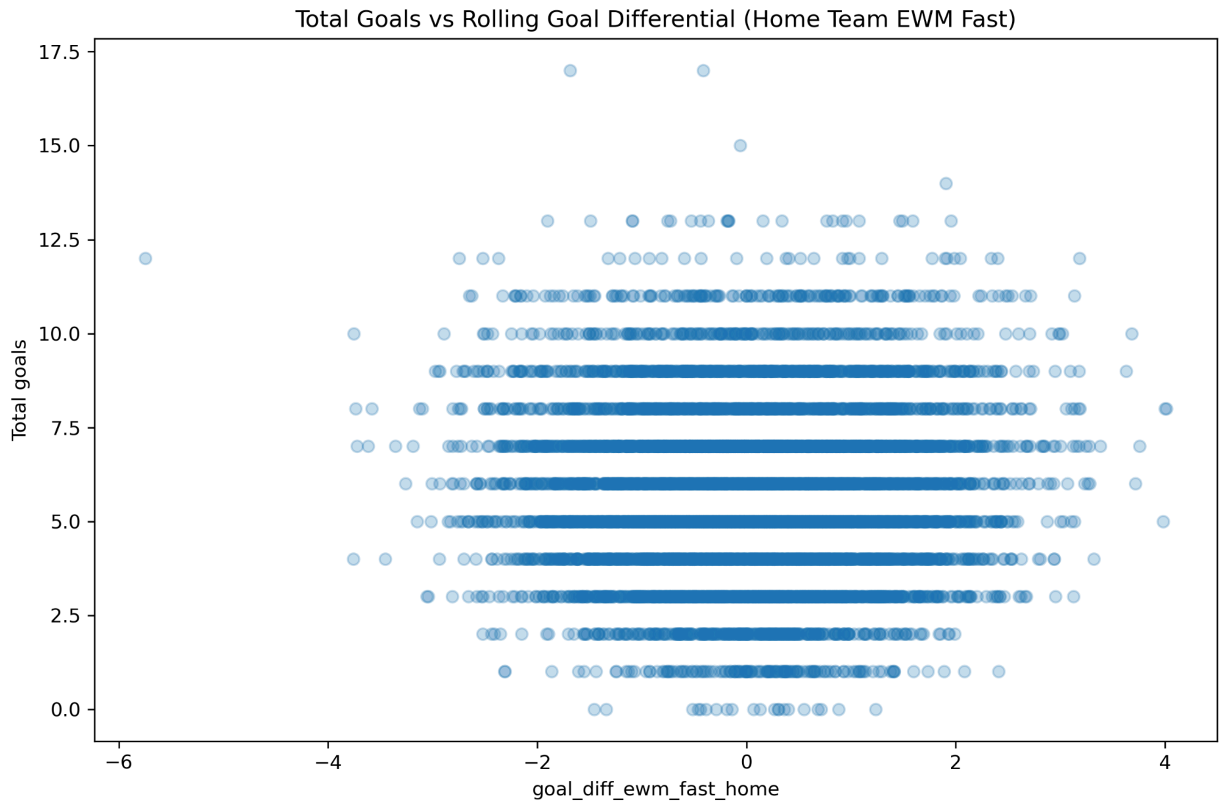

Figure 4. Total goals versus rolling goal differential (home team, fast EWM).

Figure 4 reveals a dense, roughly circular cloud centered near zero goal differential, with lighter outer rings corresponding to more extreme mismatches. The correlation between rolling goal differential and total goals is extremely small (r = 0.02), indicating that team strength alone explains little of the variation in scoring environments. This finding supports the inclusion of pace and shot-quality metrics rather than relying solely on outcome-based indicators.

3.5 Correlation Structure of Key Features

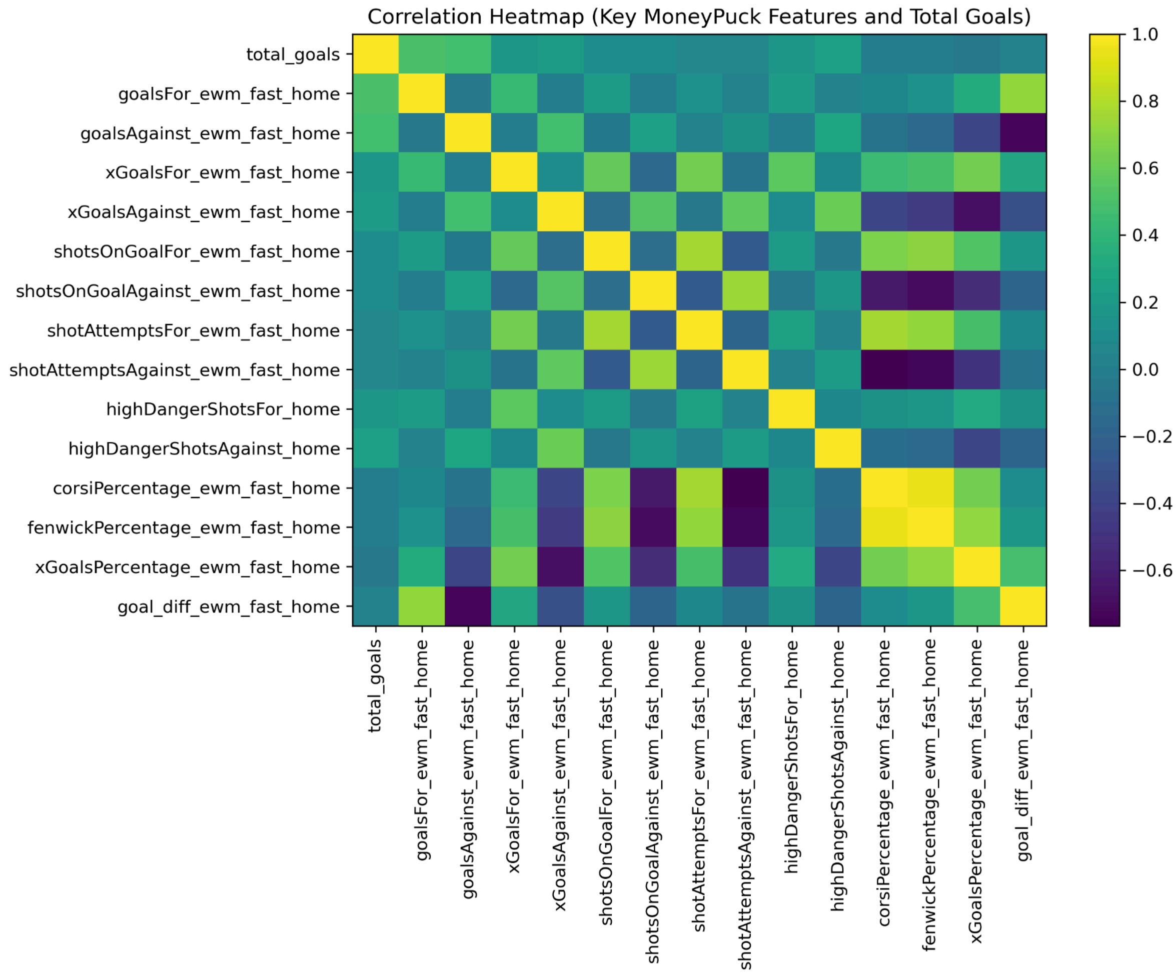

Figure 5. Correlation heatmap of selected MoneyPuck-derived features and total goals.

Figure 5 highlights several intuitive relationships. The strongest positive correlations with total goals are observed for goalsFor_ewm_fast_home (r = 0.51) and goalsAgainst_ewm_fast_home (r = 0.48), reflecting the joint contribution of offensive production and defensive leakage. Shot-attempt volume and expected-goals measures also show moderate positive correlations, while possession-share metrics cluster tightly together, indicating redundancy among Corsi, Fenwick, and xG percentage features.

4. Models

4.1 Evaluation Design and Overfitting Control

All models were evaluated using a chronological 80/20 train–test split, with earlier games used for training and the most recent games reserved for testing. To assess stability, five-fold time-series cross-validation was performed. This approach prevents training on future data and provides a realistic estimate of out-of-sample performance.

4.2 Supervised Models

Linear Regression (Baseline and Best at This Stage)

Linear regression was selected as a baseline due to its interpretability and stability. The model achieved strong and consistent performance:

- Train: MAE 0.577, RMSE 0.728, R^2 0.899
- Test: MAE 0.606, RMSE 0.759, R^2 0.892
- Time-series CV MAE scores: [0.6108, 0.5703, 0.6122, 0.6178, 0.6067]
- Mean CV MAE: 0.6036

The train–test MAE gap is approximately 0.03, indicating strong generalization.

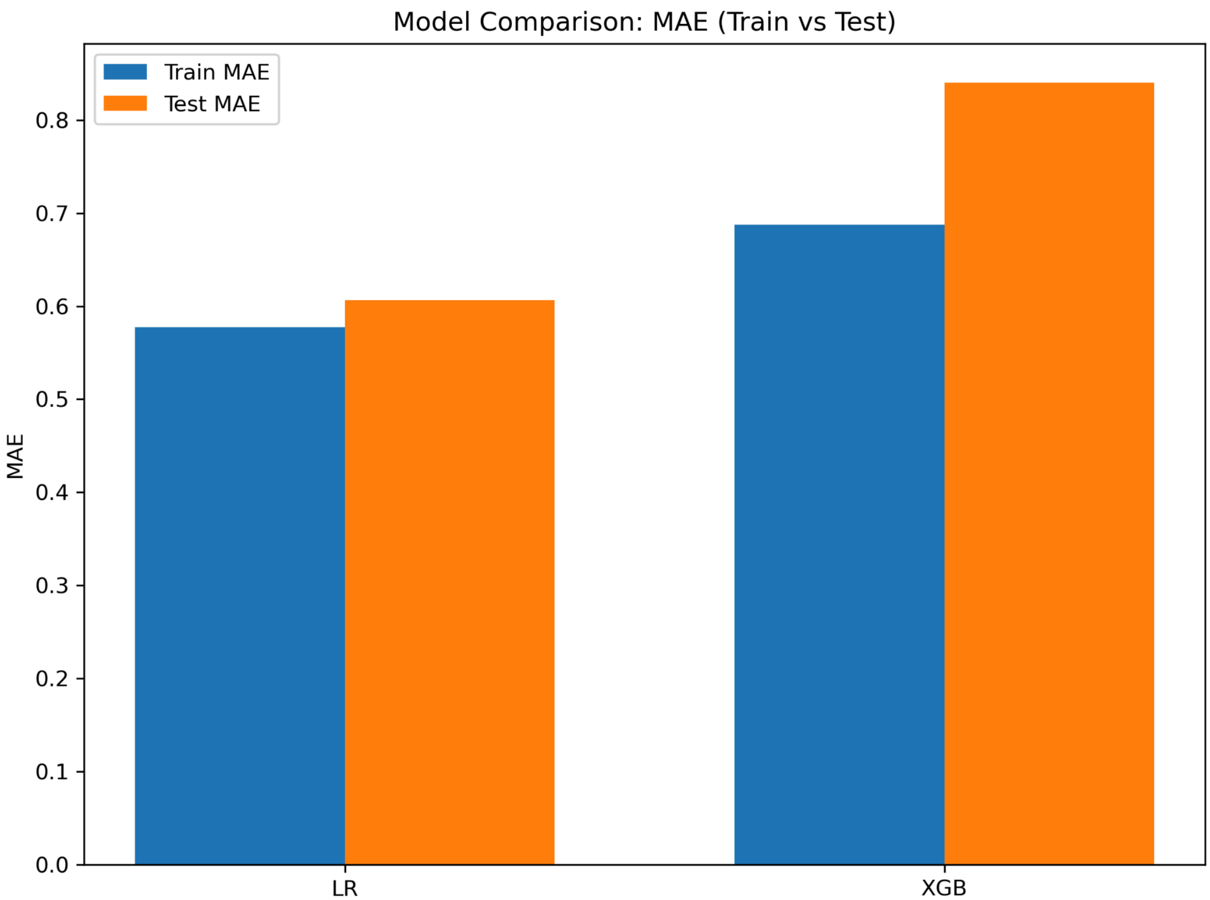

Figure 6. Train and test MAE comparison for linear regression and XGBoost.

Residual diagnostics further support model reliability. The Durbin–Watson statistic is 1.84, suggesting limited autocorrelation, while the Breusch–Pagan test indicates heteroscedasticity ($p \approx 4.1\times10^{-6}$), consistent with discrete and bounded scoring outcomes.

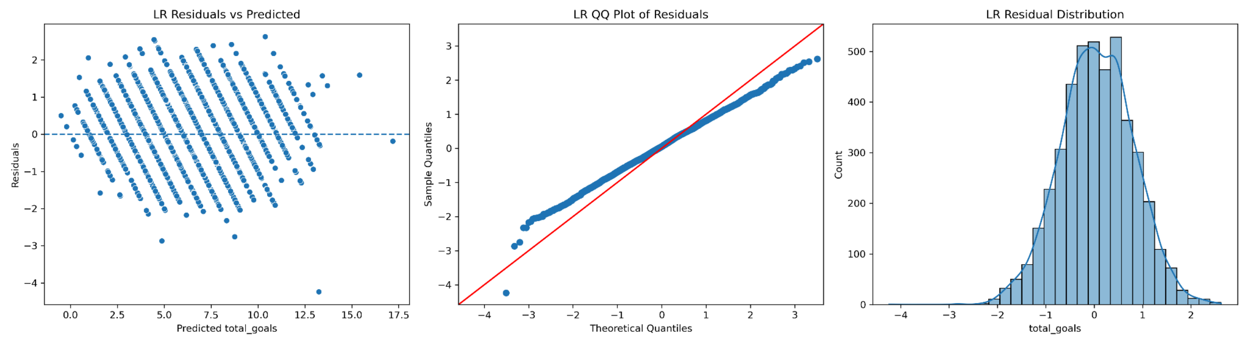

Figure 7. Linear regression residual diagnostics.

XGBoost Regression (Comparison Model)

A tuned XGBoost model was evaluated to test whether nonlinear relationships improve totals prediction. Although training performance was strong, generalization was substantially weaker:

- Test: MAE 0.840, RMSE 1.051, R^2 0.794
- Mean time-series CV MAE: 0.8417

The train–test MAE gap for XGBoost is approximately 0.15, indicating materially worse generalization than linear regression.

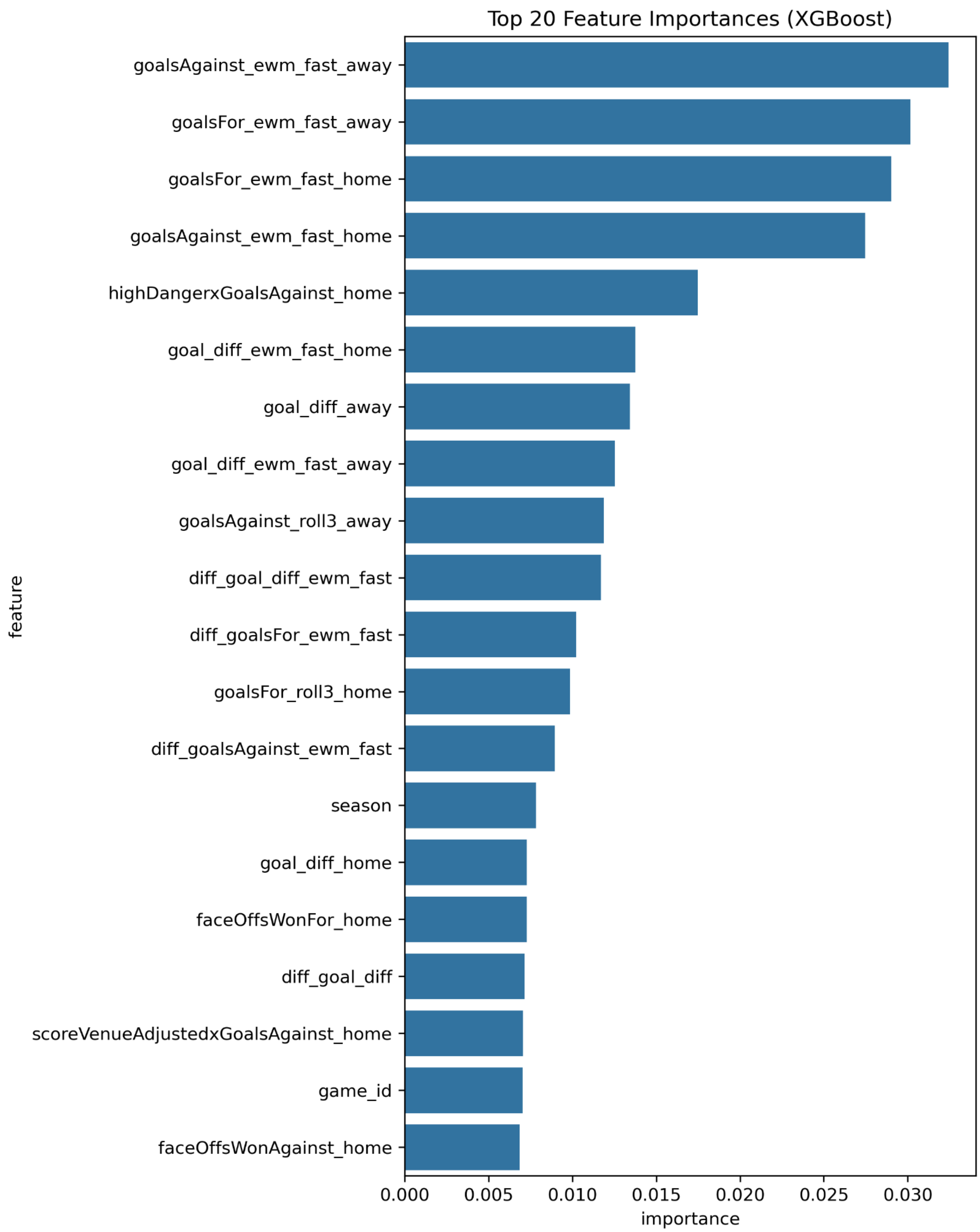

Figure 8. Top 20 XGBoost feature importances.

4.3 Unsupervised Model: Team Style Clustering

Team-level rolling features were aggregated to form a 38-team × 37-feature style matrix. K-means clustering was applied, and silhouette analysis favored k = 2 (silhouette score 0.266). Principal component analysis shows clean separation between the two clusters.

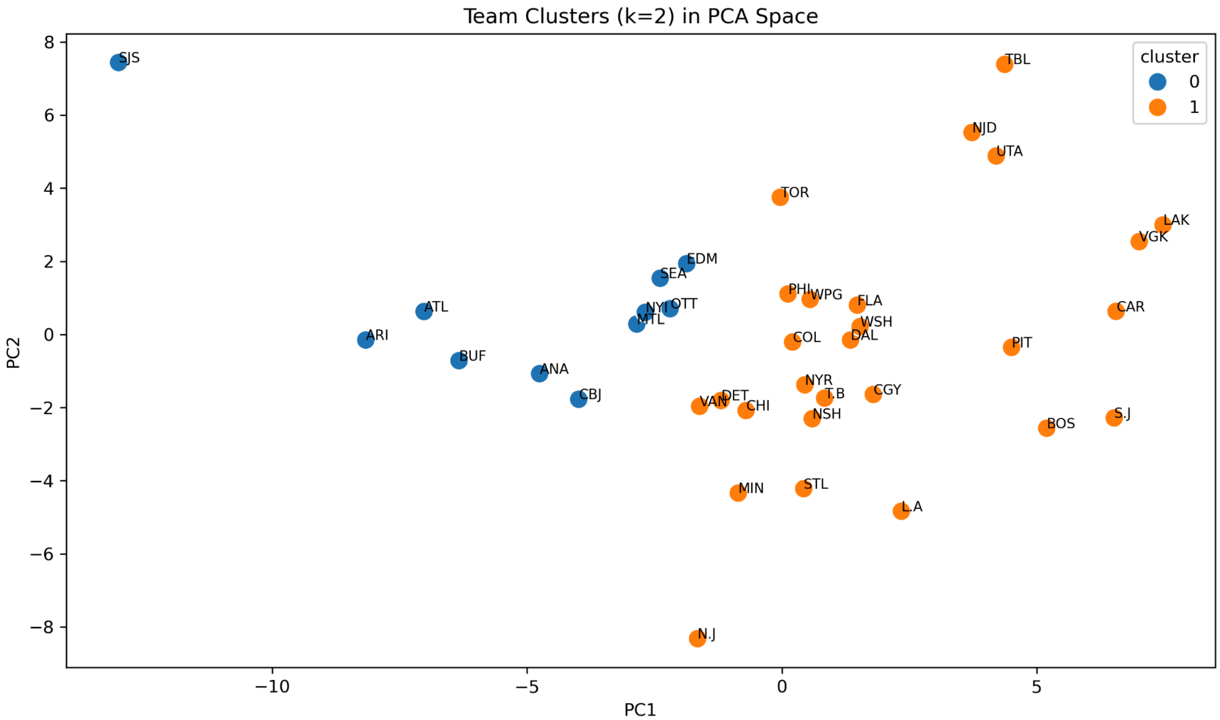

Figure 9. PCA visualization of team style clusters.

Cluster means indicate that one group exhibits higher offensive output, stronger possession and expected-goals shares, and positive goal-differential trends (high-event teams), while the other group exhibits more constrained scoring environments (low-event teams).

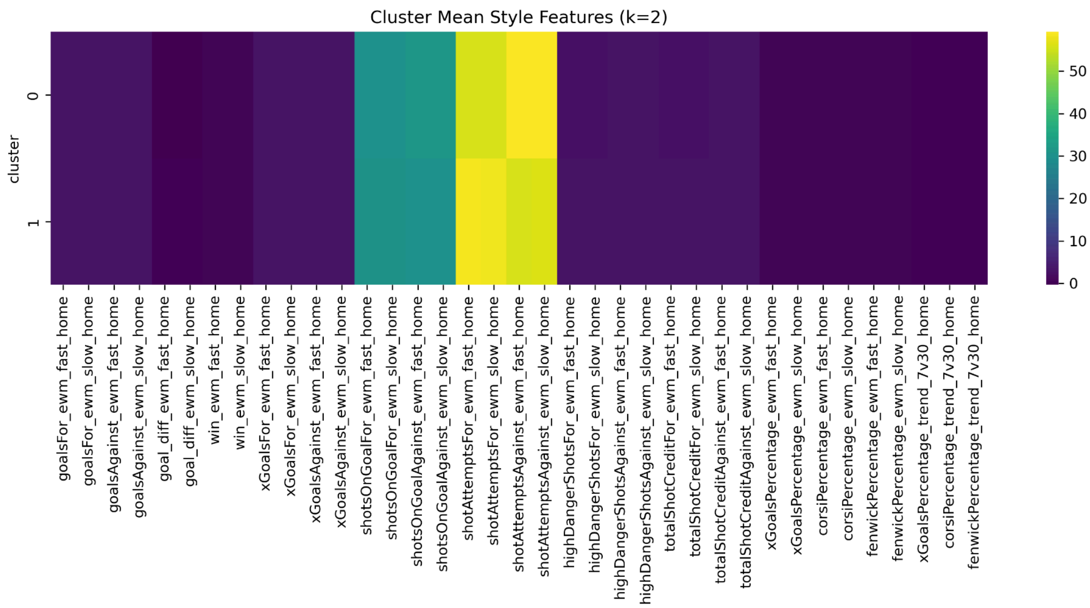

Figure 10. Heatmap of cluster mean feature profiles.

5. Discussion and Next Steps

This project demonstrates that NHL game totals reflect meaningful structure related to pace, shot quality, and team tendencies. Exploratory analysis shows substantial temporal drift in scoring levels, reinforcing the importance of time-aware evaluation. Among the models tested, linear regression provides the best balance of interpretability, stability, and out-of-sample performance, while more flexible nonlinear models introduce variance without corresponding gains in predictive accuracy.

Several limitations warrant caution. Hockey scoring is inherently noisy and discrete, leading to heteroscedastic errors that no model can fully eliminate. Important confounders such as goaltender starts, injuries, special-teams performance, and in-game tactical adjustments are not explicitly modeled. As a result, the current model should not be used for blind wagering.

Future work will focus on integrating team-style interactions into the regression framework, exploring count-aware model classes, and conducting systematic backtesting once decision rules are specified. These steps will determine whether the observed predictive structure can be translated into sustainable real-world value.

References

MoneyPuck. (2024). NHL advanced statistics and expected goals data. https://moneypuck.com

Hockey-Reference. (2024). National Hockey League scoring and historical statistics. https://www.hockey-reference.com

Appendix A — Data Dictionary (master_cleaned.csv)

A.1 Dataset overview

- Unit of observation: one NHL game (one row), containing home/away team features, engineered "recent form" features, and matchup differentials.
- Rows: 22,118 games (per your prior project outputs).
- Columns: 448 total.
- Target (response variable): total_goals.

A.2 Core identifiers and target

- game_id: Unique game identifier.
- game_date: Game date (YYYY-MM-DD).
- season: Season label (string year such as "2012", "2022").
- team_code_home: Home team abbreviation (MoneyPuck team code).
- opp_code_home: Away team abbreviation (opponent of home team, MoneyPuck code).
- total_goals: Total goals scored in the game (goalsFor_home + goalsFor_away). This is the modeling target.

A.3 Naming conventions

Team side suffixes

- *_home: feature for the home team
- *_away: feature for the away team

Rolling-window features (recent form)

- {metric}_roll{W}_{side}: rolling mean of {metric} over the last W games for that team, computed in chronological order
  - Windows used: 3, 7, 15, 30
  - Example: xGoalsFor_roll7_home

Exponentially weighted moving averages (EWMs)

- {metric}_ewm_fast_{side}: EWM with faster decay (more weight on recent games)
- {metric}_ewm_slow_{side}: EWM with slower decay (smoother, longer memory)

- Example: goalsAgainst_ewm_fast_away

Momentum / trend features

- {share_metric}_trend_7v30_{side}: short-term minus long-term form
  - Defined as: {share_metric}_roll7 − {share_metric}_roll30
  - Example: corsiPercentage_trend_7v30_home

Matchup differentials (home − away)

- diff_{feature}: home value minus away value for that same feature within a game
  - Example: diff_xGoalsAgainst_ewm_fast = xGoalsAgainst_ewm_fast_home − xGoalsAgainst_ewm_fast_away

A.4 Base MoneyPuck game-level statistics (appear as _home and _away)

These are “single-game” team stats, recorded for each team in each game, then reshaped into one row per game.

Scoring

- goalsFor: goals scored by the team in the game
- goalsAgainst: goals allowed by the team in the game
- goal_diff: goalsFor − goalsAgainst
- win: 1 if goalsFor > goalsAgainst, else 0

Shot volume (pace/possession proxies)

- shotsOnGoalFor / shotsOnGoalAgainst
- shotAttemptsFor / shotAttemptsAgainst
- blockedShotAttemptsFor / blockedShotAttemptsAgainst
- missedShotsFor / missedShotsAgainst
- totalShotCreditFor / totalShotCreditAgainst
- scoreAdjustedTotalShotCreditFor / scoreAdjustedTotalShotCreditAgainst

Chance quality

- highDangerShotsFor / highDangerShotsAgainst
- mediumDangerShotsFor / mediumDangerShotsAgainst
- lowDangerShotsFor / lowDangerShotsAgainst

Expected goals (xG) family

- xGoalsFor / xGoalsAgainst
- flurryAdjustedxGoalsFor / flurryAdjustedxGoalsAgainst
- scoreVenueAdjustedxGoalsFor / scoreVenueAdjustedxGoalsAgainst
- lowDangerxGoalsFor / lowDangerxGoalsAgainst
- mediumDangerxGoalsFor / mediumDangerxGoalsAgainst
- highDangerxGoalsFor / highDangerxGoalsAgainst

Share metrics (rate/percentage features)

- xGoalsPercentage: team share of expected goals (0–1)
- corsiPercentage: team share of shot attempts (0–1)
- fenwickPercentage: team share of unblocked shot attempts (0–1)

Physical / discipline / puck management

- penaltiesFor / penaltiesAgainst
- penalityMinutesFor / penalityMinutesAgainst
- hitsFor / hitsAgainst
- takeawaysFor / takeawaysAgainst
- giveawaysFor / giveawaysAgainst
- faceOffsWonFor / faceOffsWonAgainst

A.5 Engineered features (appear as _home and _away unless noted)

Games played index

- games_played_{side}: cumulative game count for that team within the dataset ordering (used for sequencing).

Rolling and EWM features (recent form)

For each metric in Section A.4 that is included in “metrics_for_roll” list, the dataset contains:

- metric_roll3_{side}
- metric_roll7_{side}
- metric_roll15_{side}
- metric_roll30_{side}
- metric_ewm_fast_{side}
- metric_ewm_slow_{side}

Win-rate rollups

- win_pct_roll5_{side}, win_pct_roll10_{side}, win_pct_roll20_{side}: rolling mean of win over those windows.

Trend (7 vs 30)

Computed for share metrics:

- xGoalsPercentage_trend_7v30_{side}
- corsiPercentage_trend_7v30_{side}
- fenwickPercentage_trend_7v30_{side}

Strength of schedule / opponent strength

- team_strength_xg30_home: lagged long-horizon xG% proxy (pre-game “strength” estimate).
- team_strength_xg30_opp_home: opponent’s team_strength_xg30 joined into the row (home team’s opponent strength).
- sos_xg30_roll5/10/20_{side}: rolling mean of opponent strength faced recently (schedule difficulty proxy).

A.6 Differential feature set (diff_*)

The dataset includes a wide set of matchup gaps:

- diff_{base_stat} (e.g., diff_xGoalsFor, diff_corsiPercentage)
- diff_{rolling_stat} (e.g., diff_goalsFor_roll15, diff_corsiPercentage_roll30)
- diff_{ewm_stat} (e.g., diff_xGoalsAgainst_ewm_fast)
- diff_{trend_stat} (e.g., diff_xGoalsPercentage_trend_7v30)
- diff_{sos/strength stat} (e.g., diff_sos_xg30_roll10, diff_team_strength_xg30)

Interpretation: positive values favor the home team (higher home metric), negative values favor the away team.